SONGA

INTERNAL · PRODUCT & ENGINEERING STRATEGY

What We Built, What Others Built, What to Build Next

A code-verified audit of the Songa platform, a researched comparison against how cargo and freight marketplaces actually operate elsewhere, and a proposal for what to add — including three corrections to Documents 1 and 2.

DOCUMENT	PREPARED	MARKET	BASIS
3 of 3 — Analysis	**3 August 2026**	**Tanzania · East Africa**	**Code audit + external research**

> Companion to **"Five Revenue Lines on One Codebase"** (Doc 1) and **"Tracking & Security, as a Business"** (Doc 2), both dated 20 July 2026. Every claim about our own platform is traced to a file path and was re-verified against the current tree. Every external claim carries its source and survived adversarial verification; where the evidence is thin, this document says so rather than filling the gap with plausible prose.

Executive Summary

Docs 1 and 2 argued that Songa's engine is strong enough to extend into five revenue lines and a standalone tracking business. That thesis holds — the audit confirms the expensive parts are real. But the audit also found the two documents describe an engine that differs from the one actually running, and the external research surfaced a failure mode that neither document addresses and that acts directly against Doc 1's growth sequence.

THESIS	CORRECTIONS	MISSING LINE	DEFECTS FOUND
Holds	**3**	**#0**	**6**
Redis/H3 dispatch, full cargo lifecycle, hash-chained ledger, LATRA reporting all verified in code	Doc 1 misstates the lifecycle, cites dead code, and understates what already ships	Driver retention & anti-leakage — the strongest-evidenced risk, absent from both docs	Code-verified, including two fraud detectors that can never fire

01 The engine is real, but not the one Doc 1 describes

Songa is a production intra-city cargo-trucking marketplace: pure-Redis H3 candidate selection, a load→transit→unload lifecycle, a tamper-evident double-entry ledger, and a live regulator integration. Doc 1's "~80% reusable" framing is defensible. However three of its specific claims are wrong in ways that change engineering decisions — the trip lifecycle diagram describes states the code never assigns, the "tunable dispatch waves" capability points at a dead module and at wave delays that are computed and then discarded, and the payment gateway is described as placeholder-only when a full Selcom integration ships. Section 2 documents all three.

02 Peers monetize drivers differently — and our model is the risky one

The best-evidenced finding in the external research is that driver commission causes disintermediation. On a large Asian cargo platform, introducing a 15% commission **roughly doubled measured off-platform leakage (3.92% → 7.87%)** while drivers whose fees were waived showed no change; two-thirds of a 1,971-driver sample did at least one off-app job in 137 days. Peers therefore run a **two-part tariff** — a paid subscription tier that waives commission, with commission applied only to the free tier.

Songa charges a flat 10% to every driver and has `DriverLevel` tiers 1–10 that are commented "Level Index Subscription Advantage" in the matching code but carry **no fee, no billing, no expiry and no promotion logic**. We built the skeleton of the peer model, monetized none of it, and kept the part the evidence says drives drivers off-platform.

03 Sequencing implication for Doc 1

Doc 1's order — Scheduling, then Fleet, then Corporate — is sound on effort grounds and this document does not overturn it. Two adjustments follow from the findings: the cross-cutting "platform track" is *cheaper* than Doc 1 assumed because the payment gateway is already built, and **supply retention should precede supply scaling**, because leakage compounds against every additional truck the fleet-owner line brings on.

> **How to read this document**
>
> Section 1 is what we have (code-verified). Section 2 is where Docs 1–2 need correcting. Section 3 is what others do (with explicit evidence quality). Section 4 is the proposal. Section 5 is the merged roadmap. Where research came back empty — notably African platforms and cargo trip-lifecycle norms — that is stated as a gap, not filled with inference.

1 · Where Songa Actually Stands

CODE-VERIFIED AUDIT

Every row below was read out of the current tree. This is the baseline against which both the existing proposals and the external comparison are measured.

DIMENSION	IMPLEMENTATION AS IT RUNS TODAY
Business	Intra-city cargo/truck hailing, Tanzania. Fleet is Kirikuu / Tipper / Tanker / Towing; vehicle sizes 350 kg – 60 t; paid human loaders (`number_of_couriers`). No passenger concept exists in `Trip` .
Candidate selection	Pure-Redis, zero Postgres. H3 resolution 9, k-ring expansion `[k,2,5,10,20,50,100,160]` , stops at ≥5 drivers, 30 km hard cap. Filters: verified, telemetry <300 s, car type, cargo type, back-to-back if ≤5 min from free.
Dispatch	NOT AS DOCUMENTED Simultaneous broadcast to the top 5; first driver to accept wins, resolved by a row lock. Not cascading waves — see §2.
Scoring	Base = haversine km, then rating modifier (quadratic below 4.5), level bonus −0.05×level, experience bonus, heading alignment ±0.3, fairness penalty +0.15 per recent dispatch. Rating cutoff 3.5 is a soft tier, never a hard gate.
Lifecycle	NOT AS DOCUMENTED `REQUESTED → STARTED → LOADING → EN_ROUTE_TO_DESTINATION → UNLOADING → COMPLETED` . `ACCEPTED` and `EN_ROUTE_TO_PICKUP` are declared and never assigned.
Pricing	Step-function distance/duration brackets per CarType (the containing bracket's rate applies to the whole distance, not marginal), plus loading/unloading overage × waiting rate, plus a flat per-courier fee added after the fare floor. **No weight or volume pricing.**
Surge	Demand (Postgres `REQUESTED`) ÷ supply (Redis H3 sets), clamped 1.0–2.0, locked at quote time. DISABLED BY DEFAULT
Commission	Flat **10%** on every driver. Cash trips: driver keeps money, accrues debt. Digital: auto-nets against debt. Account holds at 5,000 debt or 7 days unsettled.
Payments	MORE THAN DOCUMENTED Full Selcom gateway — RSA-signed disbursements, checkout collections, 120 s reconciliation polling. Default posture is admin-initiated payouts.
Driver tiers	`DriverLevel` 1–10. Confers *only* a −0.05×level dispatch score bonus. No billing, no fee, no expiry, no auto-promotion; the promote/demote methods have zero callers.
Fraud	Cancellation geofence 50 m + dwell 300 s, off-app collusion, passive stall. Suspension writes an append-only event with a frozen account snapshot. AUTO-SUSPEND OFF BY DEFAULT
Compliance	LATRA outbound trip reporting (OAuth2, 11 regulator-named fields) and inbound vehicle verification that **fails closed** — an unreachable regulator blocks onboarding.
Ops	Live supply/demand H3 maps, live config editing without redeploy, admin trip cancel. **No manual driver assignment** — the `TRIP_REASSIGN` audit enum exists with zero call sites.

Structural absences

No multi-stop or multi-drop. No route optimization, load consolidation or backhaul. No proof of delivery of any kind — no signature, photo, recipient identity or delivery OTP. No per-trip weight or volume declaration. No scheduled or advance booking. No B2B or contract shipper accounts. No consignee tracking link. No driver bidding. No rider location tracking. No manual dispatcher assignment.

Docs 1 and 2 already identify most of these and propose lines to close them. The audit confirms they are genuine gaps, not overlooked features — with the single exception of the payment gateway, which Doc 1 lists as a gap and which in fact ships.

2 · Corrections to Documents 1 & 2

HIGHEST-VALUE SECTION

Doc 1 states that every “Exists / Reuse” claim is traceable to a real file. Three are traceable to files that do not do what the claim says. Each was re-verified this session, and each changes an engineering or funding decision.

2.1 The trip lifecycle diagram describes states the code never assigns

Doc 1 (p.3) documents `REQUESTED→ACCEPTED→LOADING→EN_ROUTE→UNLOADING→COMPLETED`. In the running system, accepting a trip sets status directly to `STARTED`; `ACCEPTED` and `EN_ROUTE_TO_PICKUP` are declared in the model and assigned **only inside test files**.

> **Consequence — a fraud detector that can never fire**
>
> The passive-stall monitor queries trips whose status is `ACCEPTED` or `EN_ROUTE_TO_PICKUP`. Because nothing ever assigns those states, the query matches nothing and the detector has never fired in production. Any roadmap item that reasons about the pickup phase — driver-side ETA, stall detection, arrival geofencing, Doc 1's corporate proof-of-delivery — inherits this hole.

Verified: `trips/utils/accept_trip.py`, `trips/tasks/driver_stalling.py`; *grep for production assignments returns test files only.*

2.2 “Tunable dispatch waves” cites dead code, and the waves are discarded

Doc 1 (p.3) lists “H3 geo-matching, tunable dispatch waves, multi-factor scoring” as an existing capability feeding lines 2, 3 and 4, citing `utils/raw/driver_matching.py`. That module is a legacy variant and is not on the live path; the running matcher is `utils/songa_s1/suggest_driver.py`.

More consequentially, the wave mechanism itself does not operate. The matcher computes `dispatch_delay_seconds = index × 15s` for each suggested driver — and nothing ever reads it. The dispatch task iterates the matched drivers and broadcasts to all of them immediately. The real dispatch model is **simultaneous broadcast, first-accept-wins**.

> **Why this matters for the proposals**
>
> Doc 1's ride-share (#2) and corporate (#4) lines both assume a wave/cascade primitive they can tune. That primitive must be finished or deliberately deleted before those lines can lean on it. The external research does *not* settle which is correct — the one comparable source describes proximity auto-assignment but is silent on whether offers are sequential or broadcast, so this stays an internal product decision rather than a copy-the-peers question.

Verified: `utils/songa_s1/suggest_driver.py:207` *is the only write; no reader exists outside test scripts.*

2.3 The payment gateway is understated — it already ships

Doc 1 (p.3) marks “Live payment gateway (M-Pesa/Airtel/Tigo/card)” as PARTIAL — “labels + placeholder tokens only; cash-settled today” — and funds it as a cross-cutting platform track alongside line #3. In fact a complete Selcom integration ships: a gateway abstraction with RSA-signed disbursements, checkout collections, reconciliation polling every 120 seconds, and configurable payout triggers, with Selcom set as the default gateway.

> **Consequence — a funded track is largely already paid for**
>
> This is the one correction that makes the plan *cheaper*. The remaining cross-cutting work is push notifications and rider-side card enablement, not building a gateway. Budget freed here is the natural funding source for the retention work proposed in §4.

Verified: `payments/gateways/selcom_business.py`, `payments/gateways/base.py`, *default gateway* `SELCOM`.

2.4 Document 2 holds up

Doc 2's engine inventory was spot-checked and its central claims stand: month-partitioned breadcrumb history and route/ETA reconstruction both exist as described. Its thesis — that the expensive, latency-critical tracking work is done and the remainder is conventional API product build — is consistent with the audit. §3 adds external support for it.

Verified: `trips/models/TripLocationHistory.py`, `trips/models/TripActivitySummary.py`.

3 · What Others Are Doing

EXTERNAL RESEARCH

Evidence quality — read before using this section

142 candidate claims were extracted from 29 sources and adversarially verified; **13 were killed and 12 survived**. Coverage is China, the United States and academic literature only. **Zero verified claims concern African platforms** (Kobo360, Lori, Sendy, Trella) and **zero concern the 2023 Convoy and Sendy shutdowns** — no primary post-mortem survived verification. The cargo trip-lifecycle dimension (dwell time, multi-stop, proof-of-delivery, labour add-ons) produced **no verified evidence at all**. Nothing on cash settlement, driver fuel credit, cargo theft or licensing regimes. Four of the twelve surviving findings rest on a *single unrefereed working paper* with proprietary data and a co-author employed by the platform studied; those four stand or fall together.

3.1 Monetization: peers use a two-part tariff

The documented cargo-platform model is a subscription tier plus *conditional* commission: a free tier capped at **2 jobs per day** carrying 15% commission, versus roughly **CNY 399–799 per month** for unlimited jobs at **0% commission**. Only 4.2% of sampled transactions actually carried commission at all.

DIMENSION	DOCUMENTED PEER MODEL	SONGA TODAY
Commission base	Free tier only; waived for subscribers	FLAT 10% ON EVERYONE
Subscription	Monthly fee, unlimited jobs, 0% take	TIERS EXIST, NEVER BILLED
Job throttle on free tier	2 jobs/day cap drives upgrade	None — unlimited, always commissioned
Tier benefit	Fee waiver (economic)	−0.05 dispatch score per level (cosmetic)

3.2 Disintermediation is the measured failure mode — and commission causes it

Introducing a 15% driver commission across 33 cities **roughly doubled detected off-platform leakage, 3.92% → 7.87%**, while fee-waived subscribers showed no change. Cancellations among the same commissioned drivers rose 23.57% → 28.54%. Over 137 days, **two-thirds of a 1,971-driver sample transacted off-platform at least once.** Raising take-rate manufactures the leakage that erodes freight-marketplace margins.

A detector we can build with data we already have

The same source documents an implementable technique: cross-reference *cancelled* jobs against *driver* GPS traces, and flag disintermediation when the assigned driver's GPS falls near **both origin and destination** within the cancelled trip's window and the customer completed no similar on-platform trip. This needs **only driver GPS** — which Songa already stores. Our own off-app collusion detector is currently inert precisely because it was built to require rider location, which the platform does not track. This technique routes around that blocker.

3.3 Dispatch, pricing and the human desk

- **Intra-city peers auto-assign on proximity**, customer choosing vehicle class up front — no bidding, no dispatcher desk — at 400–500k matches/day with **over 20% cancellation**. Our cancellation rate is therefore probably normal, not pathological. The source is silent on sequential-vs-broadcast, so it cannot settle §2.2.
- **Freight does not surge-price.** Uber Freight prices a load as a sequence of prices across its 3-hour-to-5-day lead time, solved as a Markov decision process — not an instantaneous multiplier. Shipping our surge disabled by default was the right call.
- **A human operations desk stays in the loop** even at the most automation-forward broker: the two documented fallbacks when matching fails to cover a load are proactive human outreach and rolling the load to a later date. Songa has no manual assignment and no fallback but failing the trip.
- **Matching radius is a real decision variable** (medium confidence) — too small starves fulfilment, too large inflates pickup distance — and should vary by zone and time of day. Our per-region override hook exists and is unused.
- **H3 in freight is documented for lane and ETA work, not dispatch:** origin/destination hashed at resolution 4 into a lane code, GPS pings discretized at resolution 6, route similarity computed as hex-cell overlap across ~2M lanes. This is external support for **Doc 2**, and a concrete technique its analytics tier could adopt.

4 · What Should Be Added

PROPOSAL

Docs 1 and 2 propose six growth surfaces. This section does not replace them — it adds the line they both miss, adjusts their sequencing on the evidence, and lists the correctness work that gates them.

4.1 New line #0 — Driver retention & anti-leakage

MISSING FROM DOCS 1 & 2

This is the highest-confidence item in the document because it is the one place where the strongest external evidence lands squarely on a live configuration of ours. Three components:

✓ REUSE (ALREADY SHIPS)

- Driver GPS breadcrumb history — `TripLocationHistory`
- Cancellation records with reason + geofence data — `TripAction`
- Fraud profile counters & suspension chokepoint with frozen snapshots
- Config-driven thresholds, live-editable without redeploy
- Selcom gateway for subscription collection (§2.3)

▲ BUILD (NET-NEW)

- Leakage detector: cancelled-trip × driver-GPS correlation (§3.2)
- Leakage rate as a first-class ops metric, per driver and per corridor
- `DriverLevel` → real subscription: fee, term, expiry, purchase flow
- Commission waiver for subscribers; free-tier job throttle

The monetization half is a **product and pricing decision, not primarily an engineering one**, and should be modelled on our own numbers first — the peer figures are 2019-era, from another market, and carry the §3 caveats. The detector half can proceed immediately: it needs no pricing decision, revives a detector we already shipped, and produces the measurement any pricing decision would need.

4.2 Manual dispatcher assignment

Every peer model retains a human desk; ours has none, and our only fallback when matching fails is terminating the trip as no-driver-found. The audit enum `TRIP_REASSIGN` already exists with zero call sites. Scope: an RBAC-gated, audit-logged admin endpoint to assign or reassign a driver to a live trip. Small, and it directly reduces the funnel loss Doc 1 cites as a benefit of scheduling.

4.3 Adjustments to Doc 1's plan

DOC 1 POSITION	ADJUSTMENT
Ship #3 Scheduling first	UNCHANGED Cheapest high-value line; genuinely shapes demand.
Platform track = gateway + push	CHEAPER Gateway ships (§2.3); only push and rider card remain. Redirect freed budget to §4.1.
#1 Fleet is the supply backbone	AFTER RETENTION Leakage compounds per truck added — contain it before scaling supply.
#2 & #4 rely on tunable dispatch waves	BLOCKED Primitive does not operate (§2.2) — finish or delete first.
Surge as a lever	KEEP OFF Freight prices over lead time, not by instantaneous multiplier (§3.3).
Doc 2 tracking business	SUPPORTED Corroborated; consider H3 lane-code discretization (§3.3).

4.4 Correctness backlog — six verified defects

#	DEFECT	EFFECT	SIZE
1	Passive-stall detector queries unassigned statuses	Has never fired; pickup phase unmodelled (§2.1)	S
2	Unloading duration never measured — written to a nonexistent model field	Billed waiting fee is driver-reported, unauditable	S
3	Wave delays computed, never consumed	Dispatch model is not what the docs claim (§2.2)	S
4	Payment-method enum mismatch — `'MOBILE MONEY'` vs `'MOBILE_MONEY'`	The spaced variant fails the payment guard	S
5	Cancellation fee can never be granted — requires an always-false contact flag; no fee configured	A designed policy that never executes	S
6	Transposed default coordinate on driver location	Latitude/longitude reversed in the model default	S

5 · Merged Roadmap

Dependency-ordered, merging Doc 1's five lines, Doc 2's tracking business, and this document's additions. Sizes are engineering-relative; week ranges from Docs 1 and 2 are carried through unchanged.

#	ITEM	SOURCE	RATIONALE	SIZE
0	**Correctness backlog** (6 defects)	DOC 3	Two detectors that cannot fire and an unauditable billing input. Gates anything reasoning about the pickup phase.	S
1	**Leakage detector**	DOC 3	Only quantified failure mode in the research. Uses driver GPS we already have; produces the measurement the pricing decision needs.	M
2	**Scheduling**	DOC 1 #3	Cheapest high-value line; stub screens already placed; precursor to corporate contracts.	3–4 wk
3	**Manual dispatcher assignment**	DOC 3	Every peer retains a human desk; audit enum already exists. Cuts no-driver-found loss.	S–M
4	**Driver monetization restructure**	DOC 3	Two-part tariff vs flat 10%. Model on our own leakage data from #1 first — pricing decision, not a port of peer numbers.	L
5	**Resolve dispatch waves**	DOC 3 / DOC 1	Finish or delete. Unblocks Doc 1 lines #2 and #4, which assume the primitive works.	S
6	**Fleet management**	DOC 1 #1	Supply backbone — but sequenced after retention so leakage does not scale with the fleet.	6–8 wk
7	**Songa Track**	DOC 2	Externally corroborated; lands on fleet owners from #6 as design partners. Privacy/consent review is launch-blocking.	L
8	**B2B / Corporate cargo**	DOC 1 #4	Highest ARPU. Needs fleet supply (#6), scheduling (#2) and proof-of-delivery.	8–10 wk
9	**Ride-share / Goods selling**	DOC 1 #2, #5	Demand experiments on the same rails once the backbone exists. Ride-share additionally blocked on #5.	7–9 wk ea.

What this document deliberately does not propose

Multi-stop routing, proof-of-delivery, weight/volume pricing and scheduled-booking mechanics are real absences, and Docs 1 and 2 already scope several of them. This document does **not** add independent recommendations on their design, because the research returned no verified evidence on cargo trip-lifecycle norms — prioritising them on that basis would be guesswork dressed as analysis. They remain Doc 1's proposals, on Doc 1's reasoning.

> **Recommended follow-up research**
>
> Two gaps are directly decision-relevant and both came back empty: **(a)** African cargo platform operating models — dispatch style, cash settlement, driver fuel credit and financing, licensing regimes; and **(b)** cargo trip-lifecycle mechanics — dwell time, multi-stop sequencing, waiting-fee triggers, proof-of-delivery capture, labour add-ons. Closing (a) would tell us whether the China/US monetization evidence transfers to Tanzania at all; closing (b) would let §4 speak to Doc 1's lines #2, #4 and #5 on evidence rather than deferring to them.

What this document is, and is not

It is an audit and a comparison. Claims about our platform were read out of the current tree and re-verified; claims about other platforms carry sources and survived adversarial verification, with the coverage gaps stated in §3 rather than smoothed over. It is not a market sizing or a financial forecast, and the effort ratings are engineering-relative. The monetization proposal in §4.1 in particular needs our own leakage and driver-economics numbers before it becomes a plan.